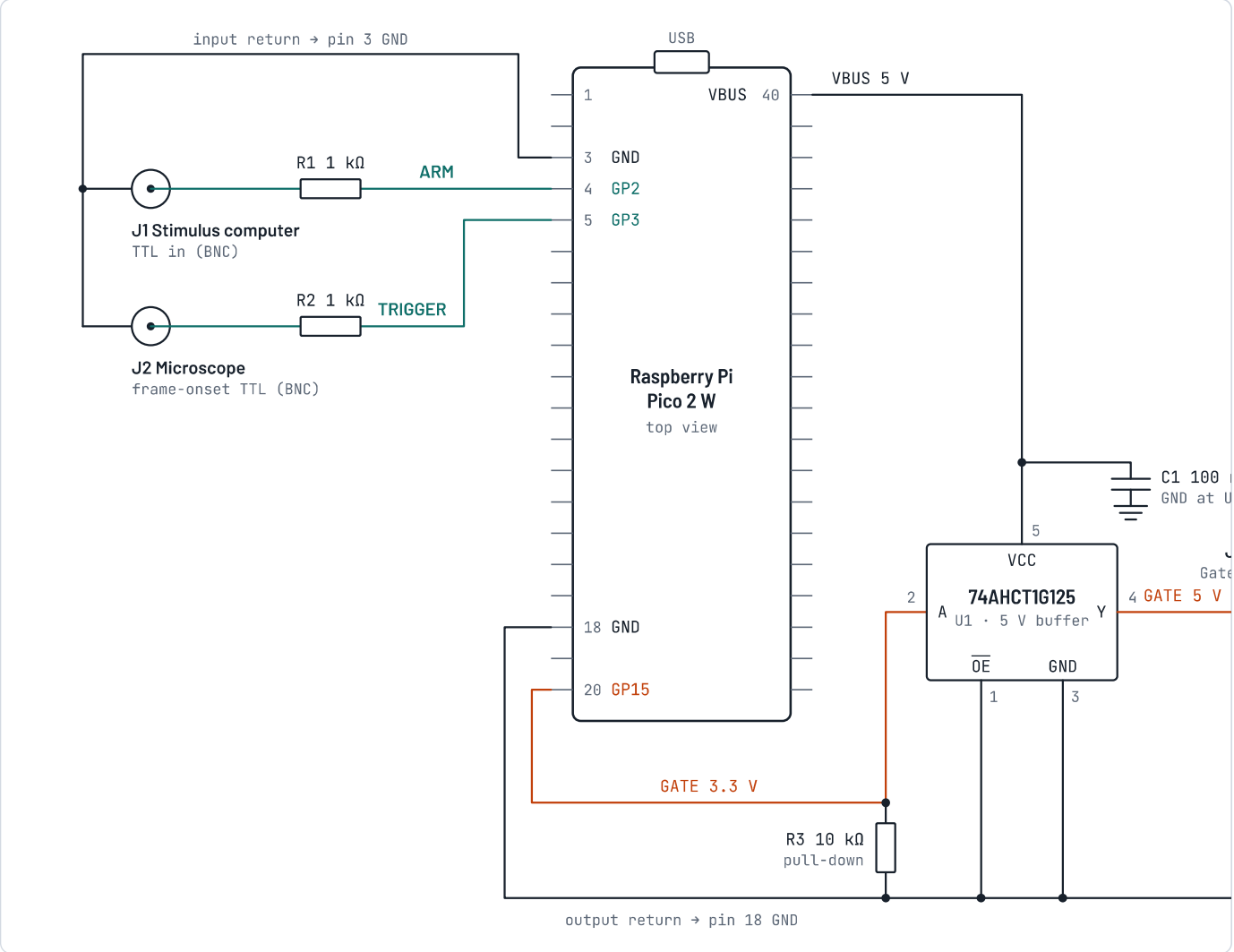
OAMCONTROL · PICO 2 W · HARDWARE

Connection schematic

TTL inputs through series resistors, gate output through a 5 V buffer to Monaco Gate 1.
Pin numbers are physical header pins; GPn are the GPIO numbers used in the firmware.



Both TTL inputs enter through 1 kΩ series resistors and return to GND pin 3. GP15 drives the 74AHCT1G125 buffer, which lifts the gate to 5 V for Monaco Gate 1. The 10 kΩ pull-down keeps the gate low while the Pico boots or is flashed, and the buffer and laser ground return to GND pin 18.

Connection list

FROM	VIA	TO
J1 stimulus TTL, centre	R1 1 kΩ	pin 4 · GP2 (ARM)
J2 microscope TTL, centre	R2 1 kΩ	pin 5 · GP3 (TRIGGER)

J1, J2 shields	–	pin 3 · GND
pin 20 · GP15 (GATE)	–	U1 pin 2 (A)
U1 pin 2 (A)	R3 10 kΩ	GND
pin 40 · VBUS	–	U1 pin 5 (VCC)
U1 pin 5 (VCC)	C1 100 nF	U1 pin 3 (GND), close to U1
U1 pin 1 ($\overline{OE}$)	–	GND (output always enabled)
U1 pin 3 (GND)	–	pin 18 · GND
U1 pin 4 (Y)	–	J3 Monaco Gate 1, centre
J3 shield	–	pin 18 · GND

Parts

REF	PART
R1, R2	1 kΩ, ¼ W
R3	10 kΩ, ¼ W
C1	100 nF ceramic
U1	74AHCT1G125, SOT-23-5 (DBV) on a breakout board
J1, J2	BNC panel sockets
J3	BNC socket, cable to Monaco HD-BNC Gate 1

Notes

VBUS is 5 V only on USB power.

If the Pico is powered another way, feed U1 VCC from that 5 V supply instead of pin 40.

U1 pin numbers are for the SOT-23-5 (DBV) package.

Other packages number the pins differently; check the datasheet of the part you buy.

R1 and R2 are the minimum input protection.

RP2350 inputs only tolerate 5 V while the Pico is powered. The resistors limit the current if a TTL arrives while it is off. The firmware's internal pull-downs keep unconnected inputs low.